

EMINES

School of Industrial Management

Plateforme Mobile

soutenance 4

SOMMAIRE

PARTIE MECANIQUE

- ▮ la maquette numérique
- ▮ Pièces fabriqués et pièces en cours de fabrication
- ▮ Conception de la pièce à imprimer en 3D

PARTIE ELECTRIQUE ET INFORMATIQUE

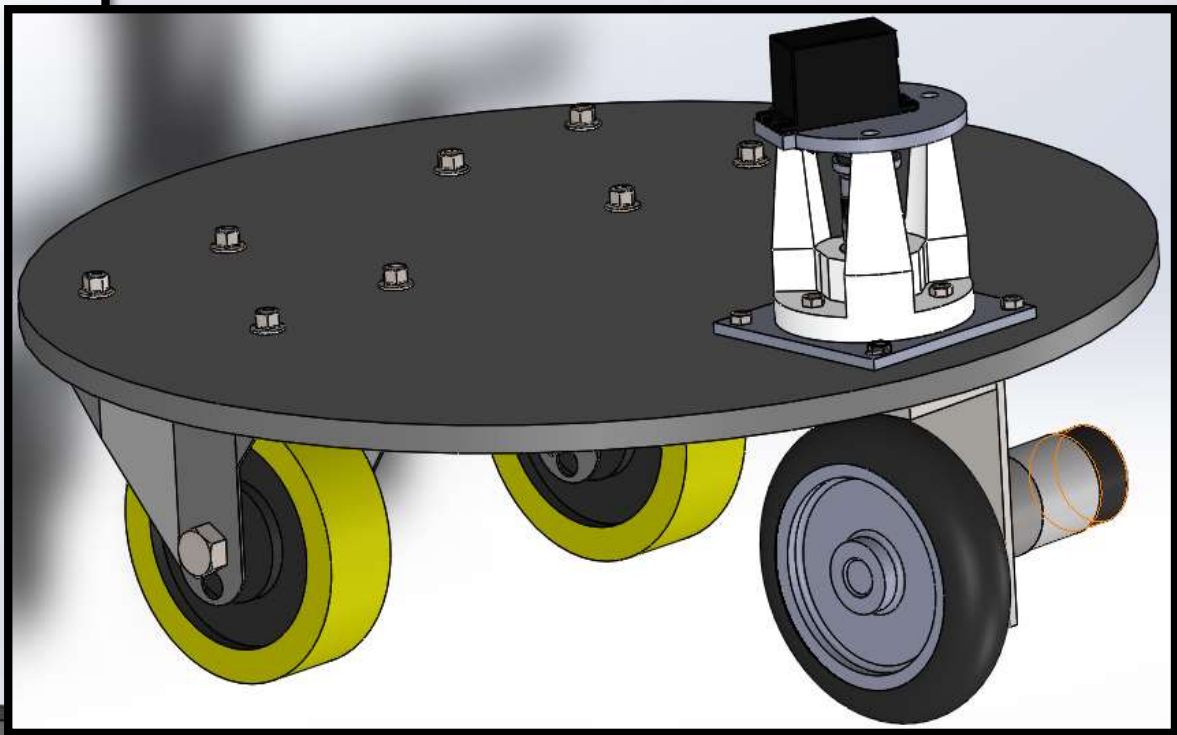
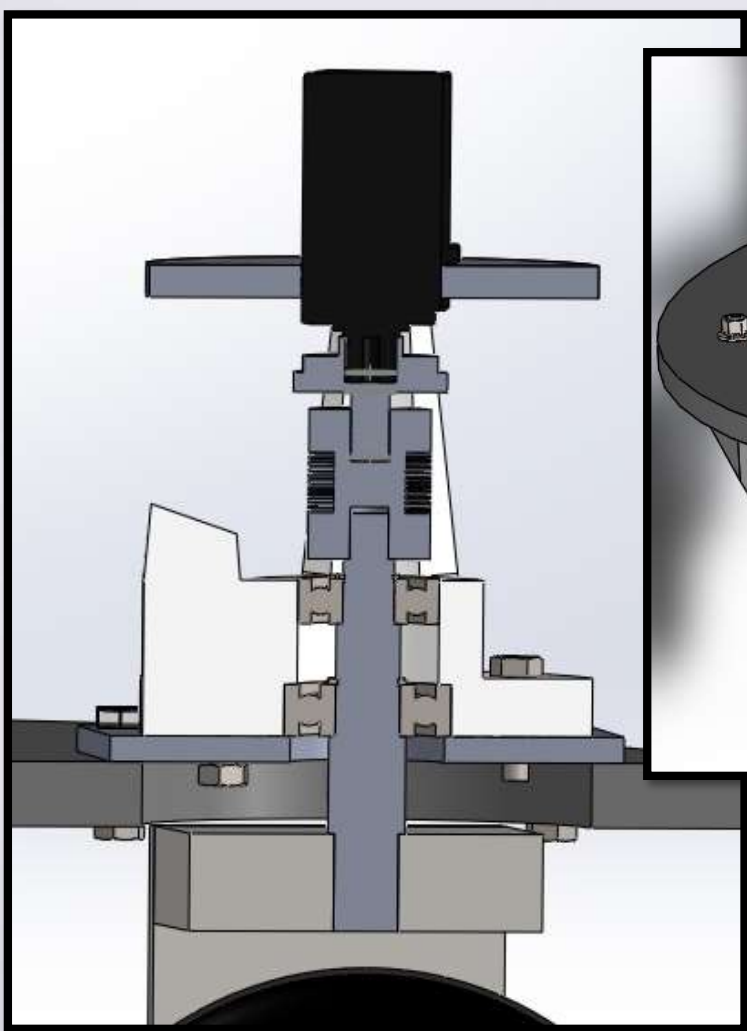
- ▮ Modélisation de la MCC
- ▮ Réponse de la MCC
- ▮ Introduction de Bloc PID
- ▮ Code Arduino et démonstration de l'asservissement en vitesse
- ▮ Interface graphique

I.

PARTIE MECANIQUE

I.

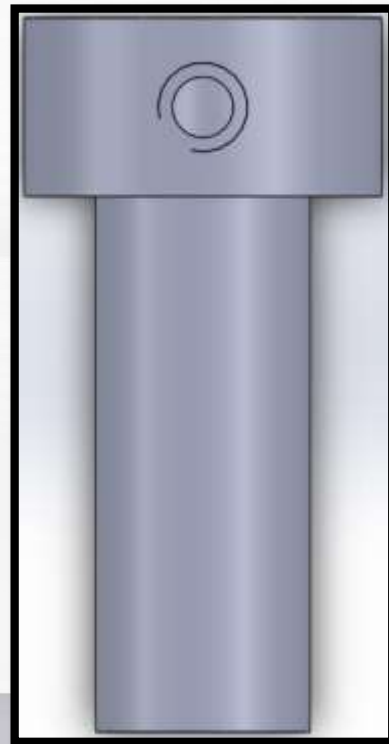
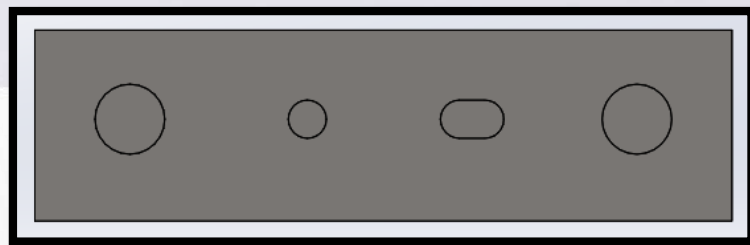
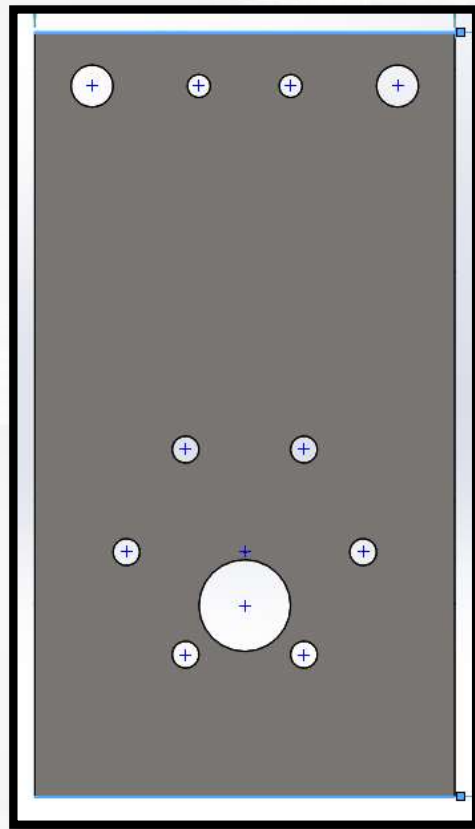
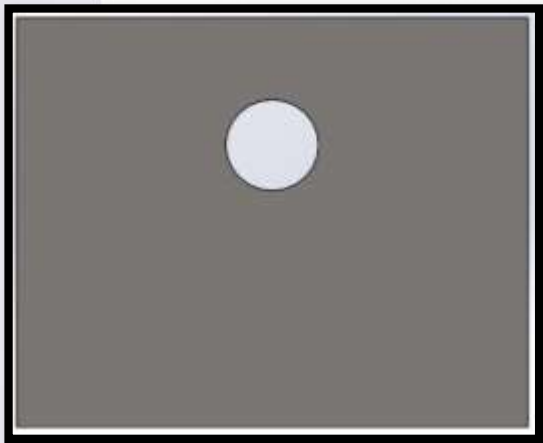
la maquette numérique



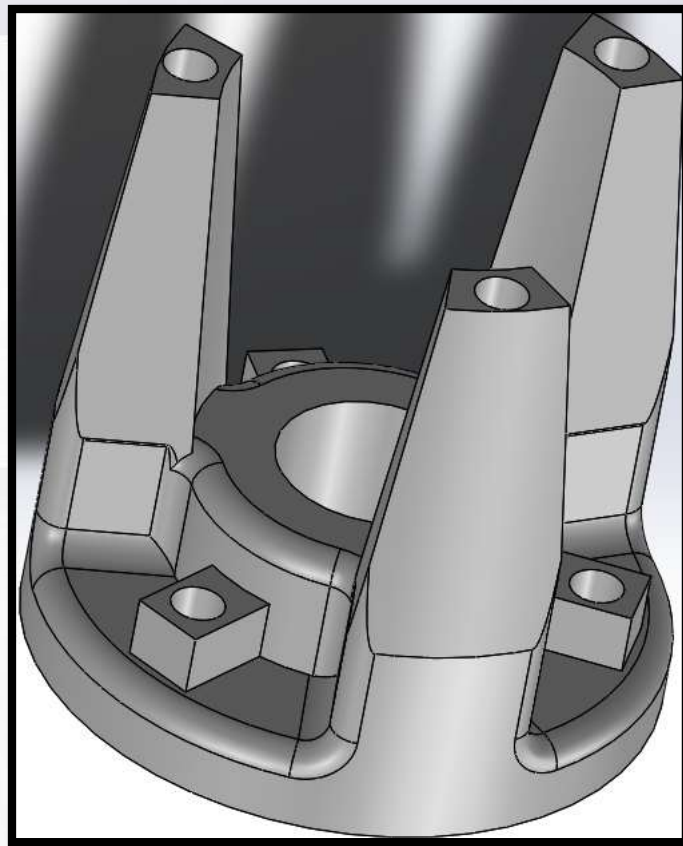
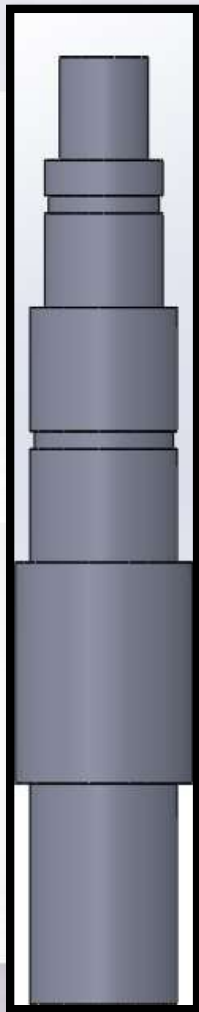
2.

Pièces fabriqués et pièces en
cours de fabrication

Pièces fabriqués

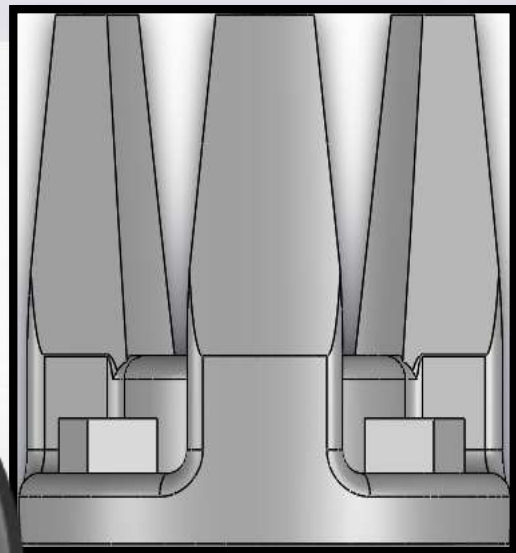
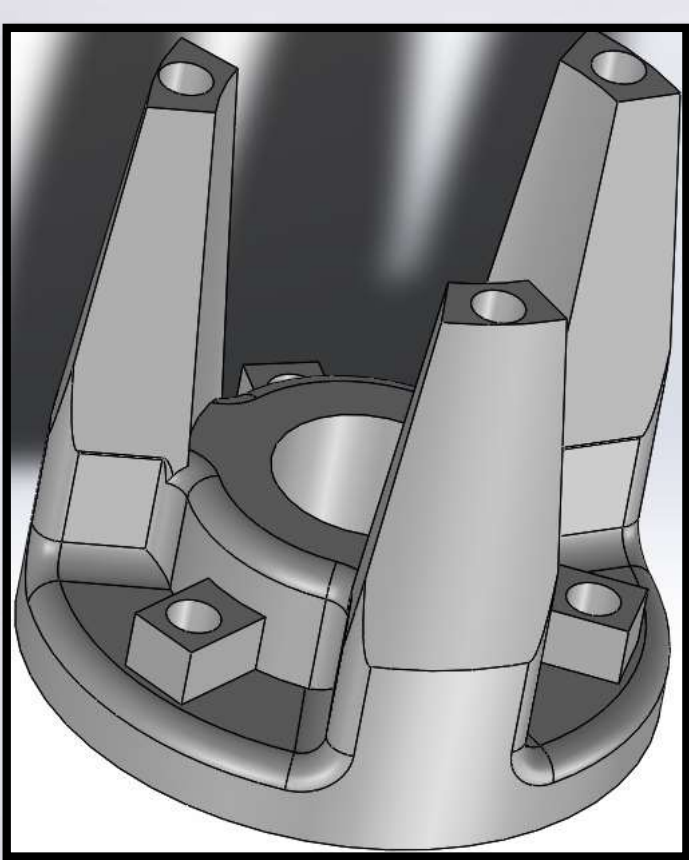


Pièces non fabriqués



3.

Conception de la pièce à imprimer en 3D

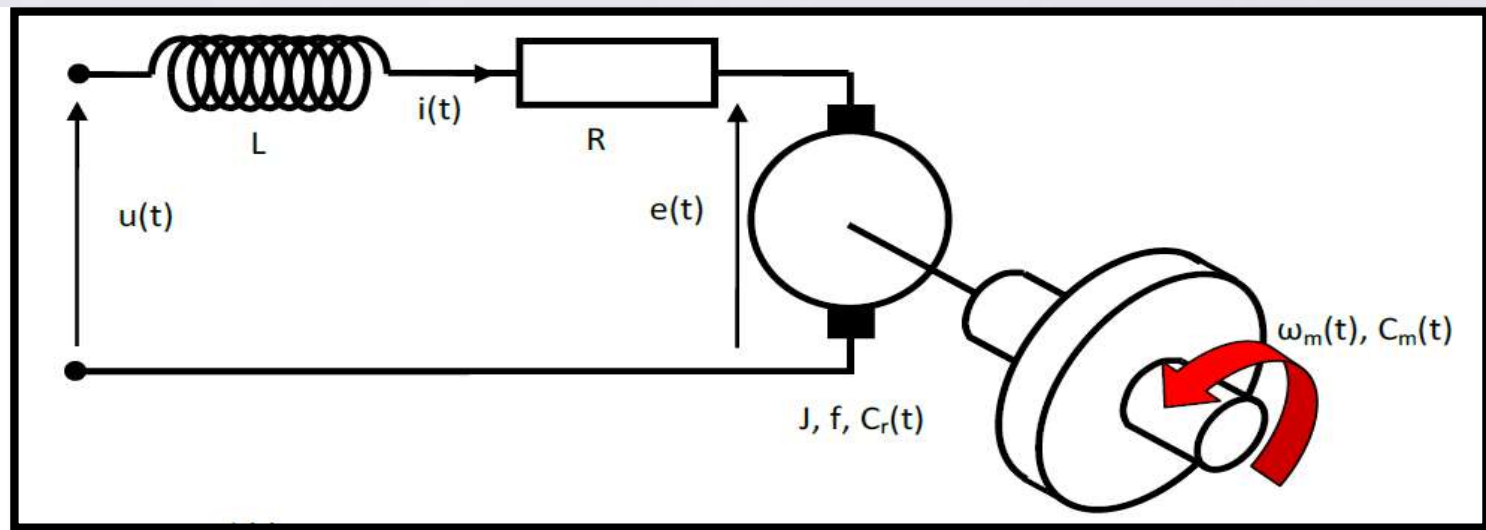


2.

PARTIE ELECTRIQUE ET INFORMATIQUE

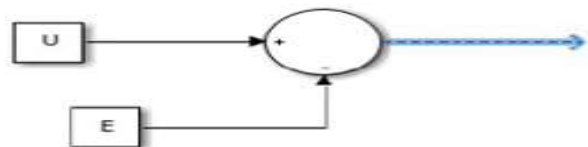
I.

Modélisation de la MCC

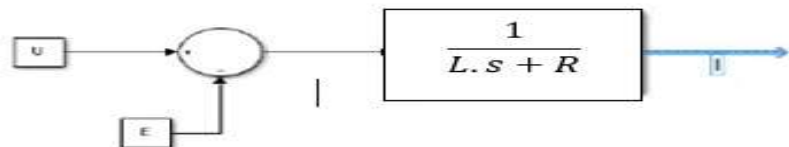


$$\begin{cases}
 u(t) = e(t) + Ri(t) + L \frac{di(t)}{dt} \\
 J \frac{d\omega(t)}{dt} = C_{em}(t) - C_r(t) - f\omega(t) \\
 e(t) = K_e \omega(t) \\
 C_{em}(t) = K_{em} i(t)
 \end{cases}
 \xrightarrow{\text{Par transformation de Laplace}}
 \begin{cases}
 \frac{I(p)}{U(p) - E(p)} = \frac{1}{R + Lp} \\
 \frac{\Omega(p)}{C_{em}(p) - C_r(p)} = \frac{1}{f + Jp} \\
 E(p) = K_e \Omega(p) \\
 C_{em}(p) = K_{em} I(p)
 \end{cases}$$

$$U - E$$



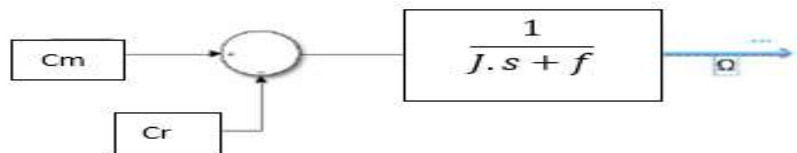
$$(U - E) \frac{1}{Ls + R} = I$$



$$E = K_e \Omega$$



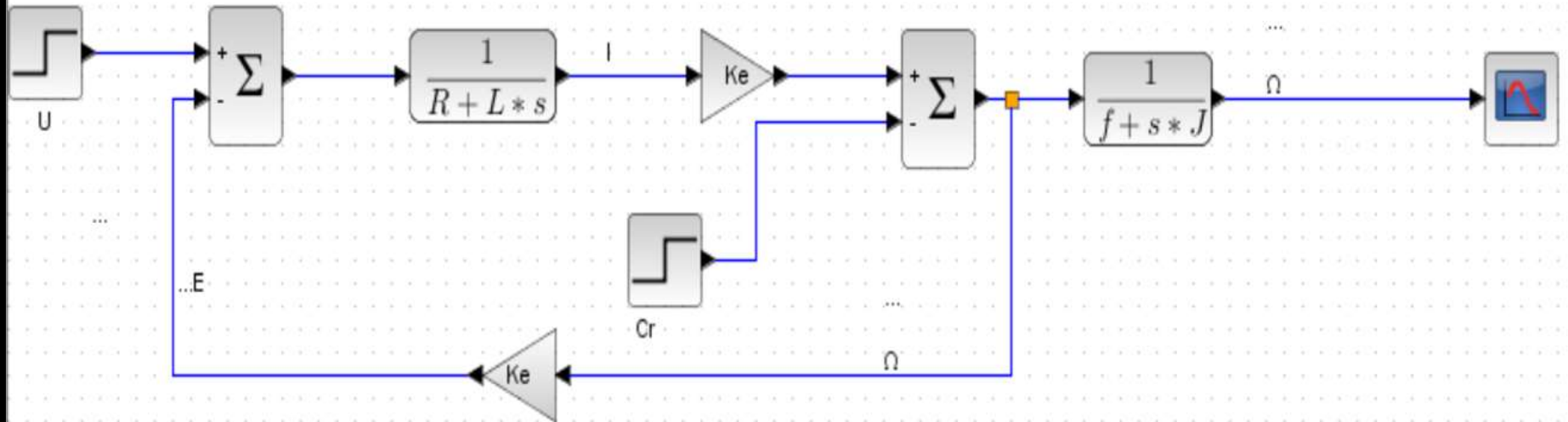
$$(C_m - C_r) \frac{1}{Js + f} = \Omega$$



$$(C_m) = K_m I$$

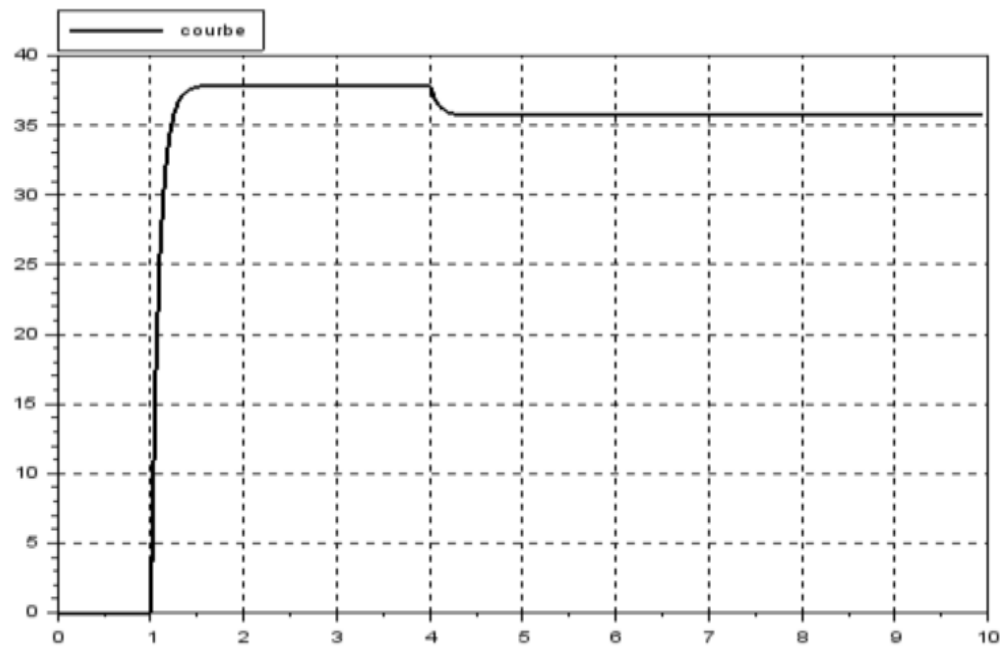


Assemblage des différents blocs précédents sur Scilab:



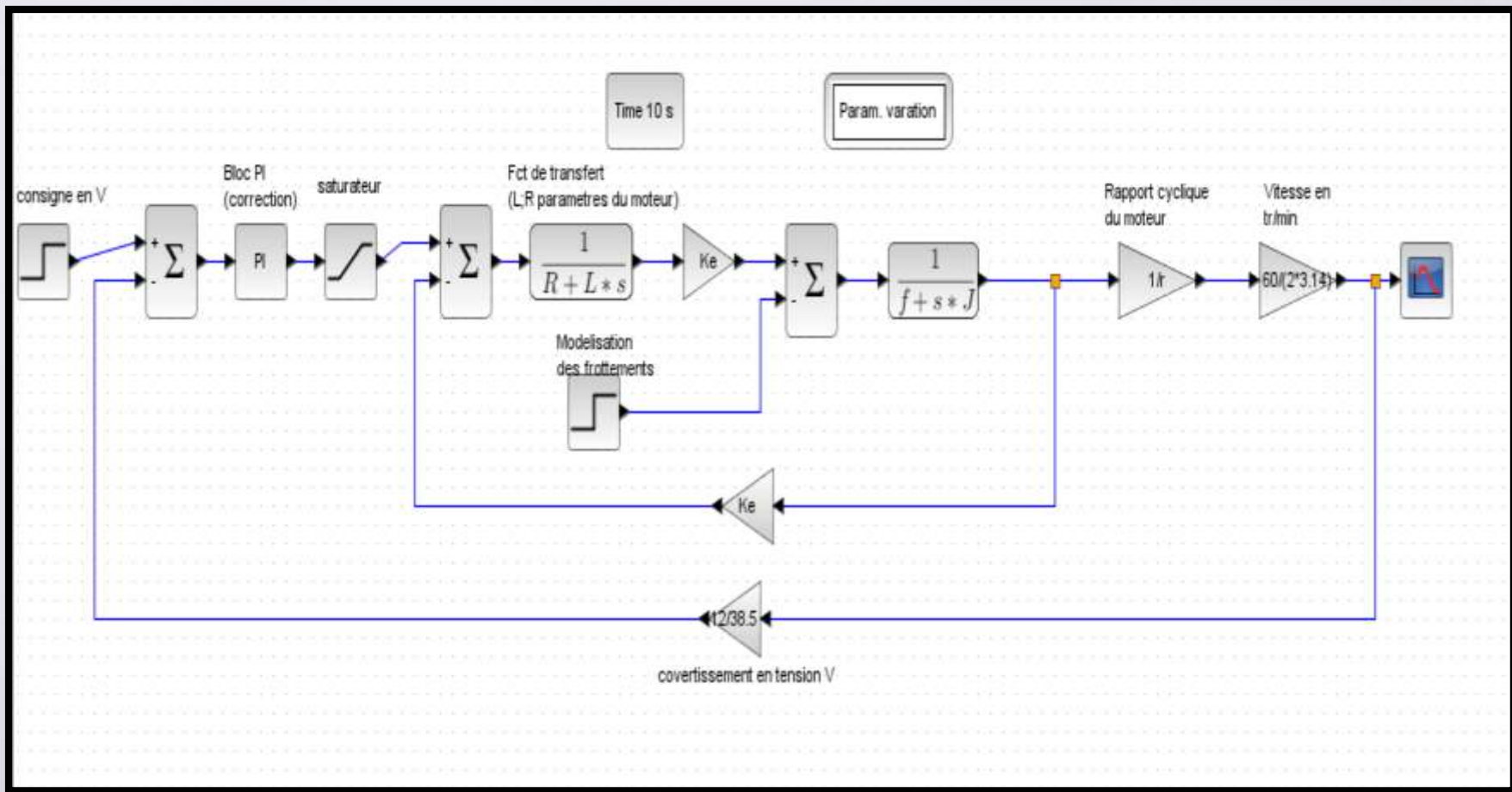
2.

Réponse de la MCC

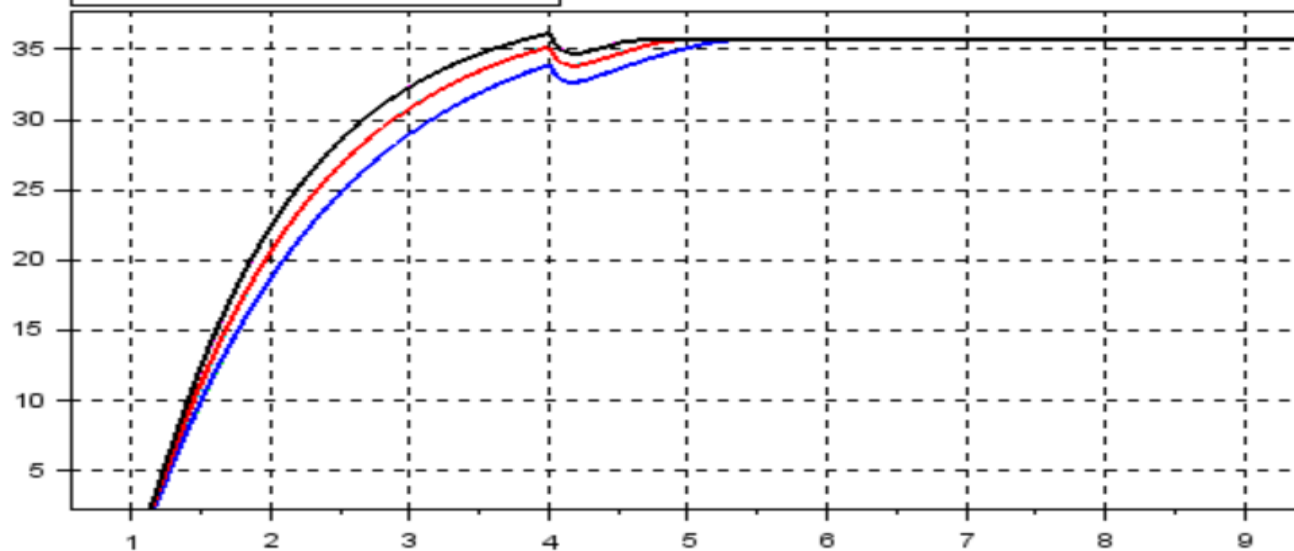


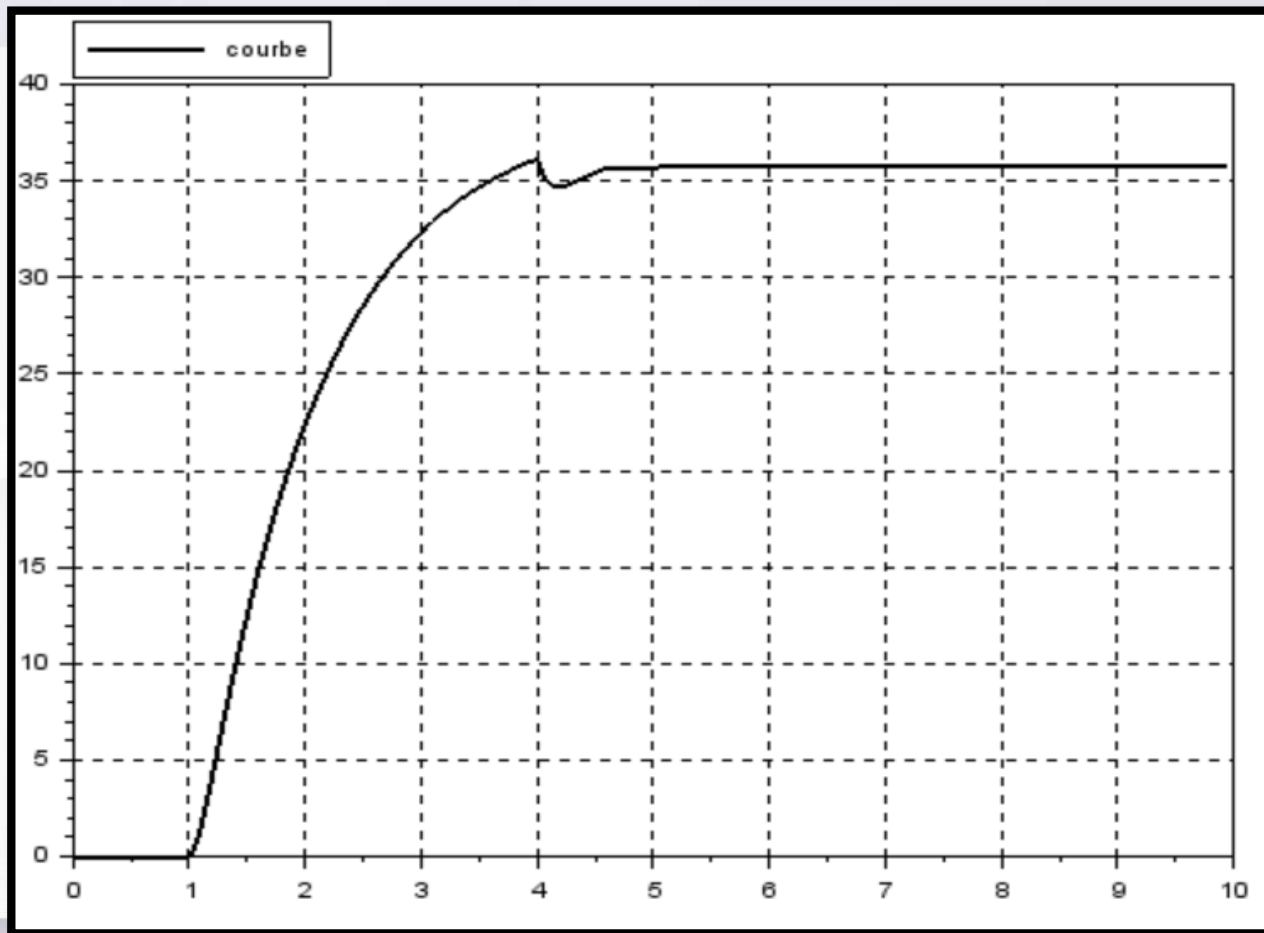
3.

Introduction de Bloc PID



- $k_p=0.003$, $k_i=0.7$ courbe
- $k_p=0.003$, $k_i=0.8$ courbe
- $k_p=0.003$, $k_i=0.9$ courbe
- $k_p=0.004$, $k_i=0.7$ courbe
- $k_p=0.004$, $k_i=0.8$ courbe
- $k_p=0.004$, $k_i=0.9$ courbe
- $k_p=0.005$, $k_i=0.7$ courbe
- $k_p=0.005$, $k_i=0.8$ courbe
- $k_p=0.005$, $k_i=0.9$ courbe

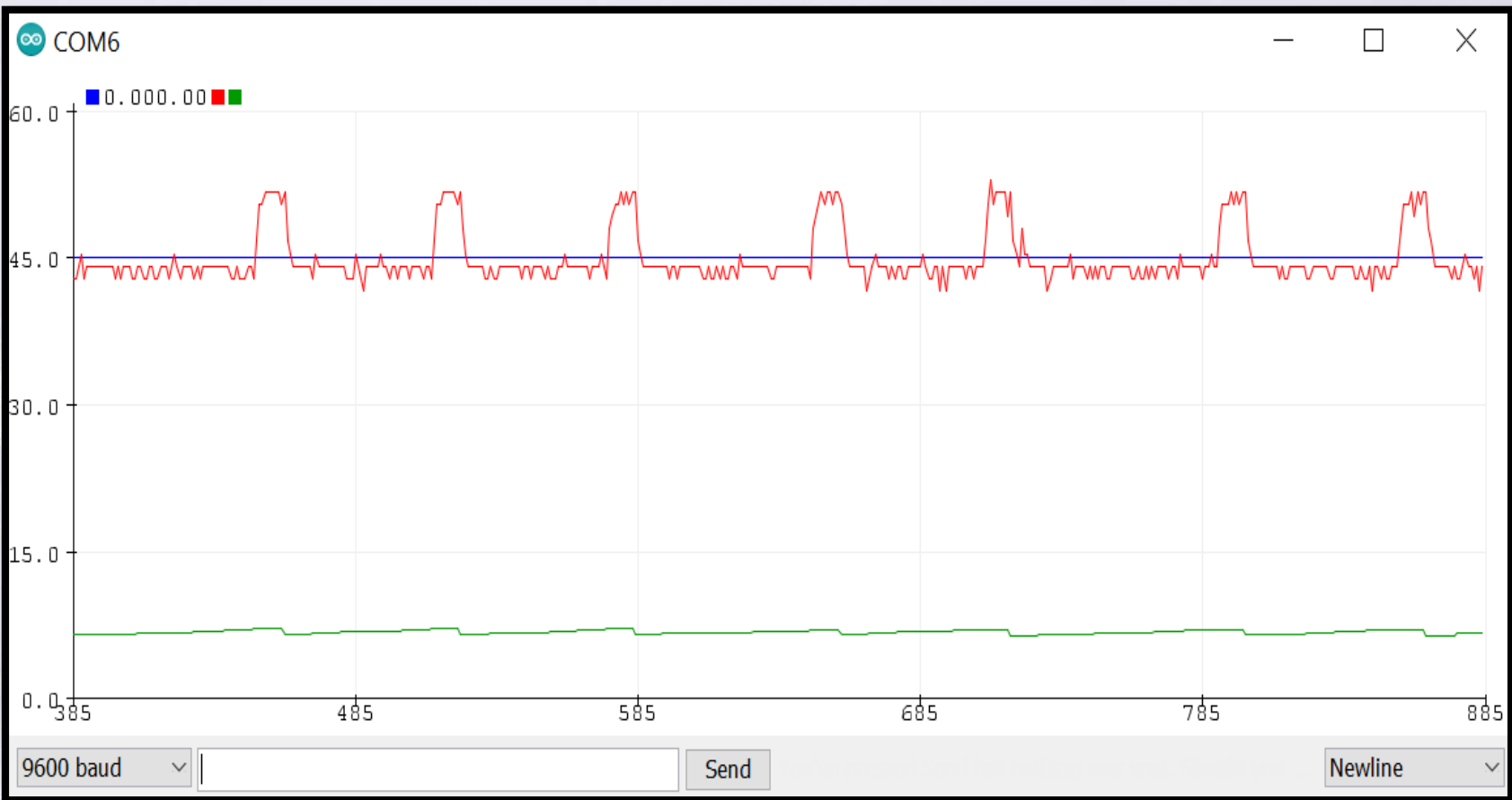




4.

Code Arduino et démonstration de l'asservissement en vitesse





5.

Interface graphique

GRUPE 7

ALL_LEFT

FORWARD

ALL_RIGHT

LEFT

STOP

RIGHT

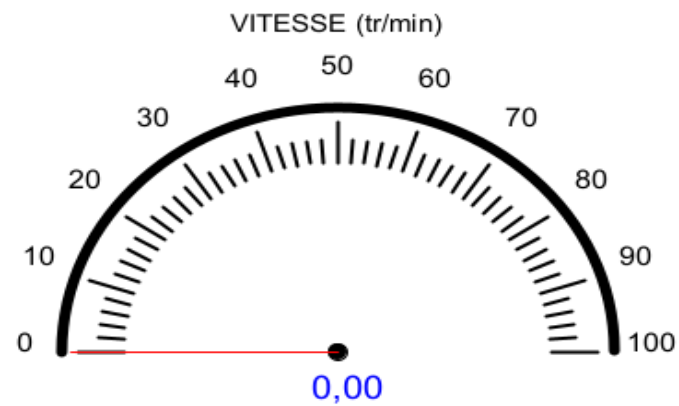
BACKWARD

SERVO_90

VITESSE

APPLIQUER

VITESSE (tr/min)



GROUPE 7

ALL_LEFT

ALL_RIGHT

LEFT

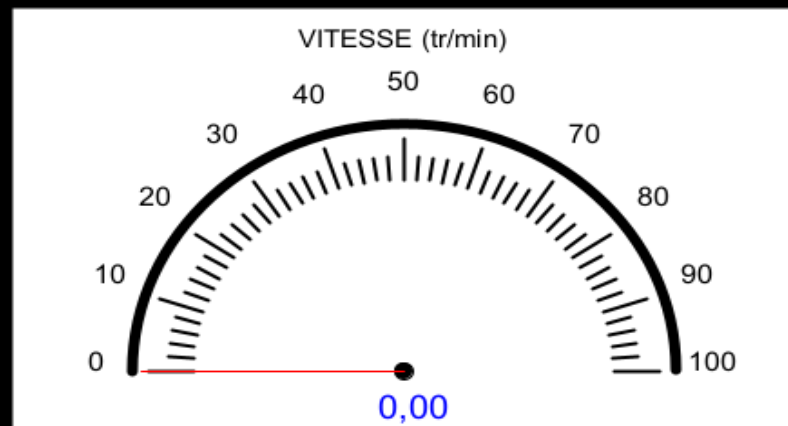
RIGHT

SERVO_90

VITESSE

APPLIQUER

VITESSE (tr/min)





MERCI POUR VOTRE
ATTENTION !